

YAHIA Omar

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Summary

Computer Systems and Networks graduate from Saad Dahleb University with experience in data engineering, cloud infrastructure, and machine learning. Skilled in Python, Docker, Kubernetes, and transforming industrial data into scalable, production-ready solutions. Passionate about building efficient data pipelines and cloud-based systems.

EDUCATION

Saad dahleb University, Faculty of Science

Blida, Algeria

Bachelor's in computer systems

june 2024

- Relevant Coursework: Machine Learning, Probability & Statistics, Databases, Algorithms, Software Engineering, Design & Analysis of Experiments, Survey Methodology

Saad dahleb University, Faculty of Science

Blida, Algeria

Master in computer systems and Network

june 2026

- Relevant Coursework: Deep Learning, Data Science, Computer Architecture, Distributed Systems, IP Networks (LAN/WAN), Network Security, Data Virtualization

PROFESSIONAL EXPERIENCE

Groupement berkine (sonatrach - oxy)

Hassi massoud, Algeria

Internship

april 2026

- Solved real-world problems by applying practical data science techniques
- Utilized machine learning methods such as feature engineering, modeling, evaluation, and testing
- Built an anomaly detection system for OT networks using federated learning and Software-Defined Networking (SDN)
- Deployed models with 93% accuracy as web applications using Scapy, Python, and Scikit-learn
- Collected, cleaned, and prepared industrial network data (Modbus, DNP3, IEC 104)

VOLUNTEER EXPERIENCE

CSCC (computer science community club)

Blida, Algeria

Dev Member / podcast Leader / vice president / president

October 2021 – June 2026

- Developed technical skills in web development, Linux, ethical hacking, data collection, and computer vision
- Contributed to data-driven decision-making within the organization
- Collaborated with team members to solve technical and organizational problems
- Prepared, organized, and participated in events such as hackathons, CTFs, and conferences

SKILLS

- **Programming Languages:** Python , Java, Ruby, shell ,SQL
- **Data Science & Miscellaneous Technologies:** Data pipelines (cleansing, wrangling, visualization, modeling, interpretation), Statistics, Time Series, Experimental Design, Hypothesis Testing, OOP, APIs, Excel, Git
- **Big Data & Machine Learning:** Docker, Spark, AWS (EC2), azure , huawei cloud, Heroku, Python (e.g.: - Scikit-learn, NumPy, Pandas, Matplotlib, Plotly, Streamlit), YOLO
- **CI/CD & Devops :** linux , kubernetes , jenkins , ansible , virtualization
- **Network :** CCNA, HCIA Cloud, Routing, Firewalls, VLANs, ACLs

HONORS AND AWARDS

Algeria 2.0

Winner of CropVision DZ track

december - 2025

- **Achieved 3rd place** by developing a solution for early detection of plant diseases and stress using AI, computer vision, and Flutter for mobile applications.

PROJECTS

Data Analysis, Visualization & Machine Learning

Fire Detection from Satellite Images

june - 2024

- Used YOLOv9 with Python to detect wildfire-prone zones from satellite imagery
- Preprocessed images, trained detection models, and visualized results
- Worked on data collection, annotation, and inference pipeline development

Citus Multi-Node Setup on Docker VPSs

july - 2025

- Designed and deployed a scalable distributed PostgreSQL cluster using Docker Compose
- Integrated high availability and replication for testing purposes

SDN-Based Architecture with Anomaly Detection for Hybrid Industrial Network Security

april-2026

- Built an SDN-based architecture integrating anomaly detection to secure industrial communications
- Used ONOS and federated learning for intelligent network monitoring

Native web Development

University Laboratory Website Remake

june - 2023

- Rebuilt a university lab website using React, Express.js, and Tailwind CSS
- Improved UI/UX, responsiveness, and content management